Strong Exposure, Preserved Extremality, and Norm Attainment in Projective Tensor Products

Candidate partial result for arXiv:2211.13559

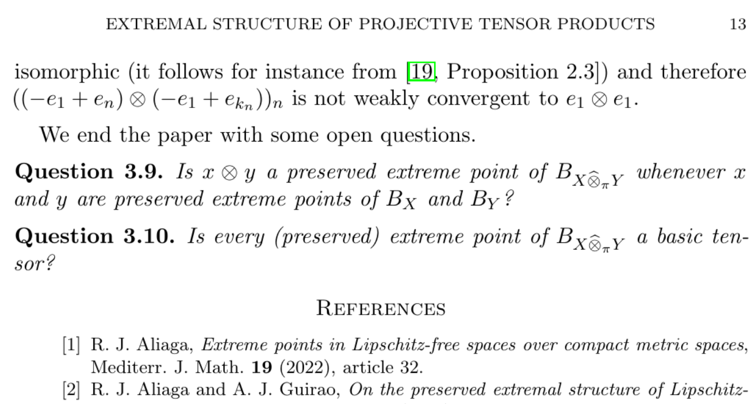
22 July 2026

Status: candidate partial result, likely valid. The two source questions remain open in full.

1 Source questions

García-Lirola, Grelier, Martínez-Cervantes, and Rueda Zoca ask the following at the end of [1] (arXiv PDF page 13).

> **Question 3.9.** Is $x \otimes y$ a preserved extreme point of $B_{X\widehat{\otimes}_\pi Y}$ whenever x and y are preserved extreme points of B_X and B_Y?
>
> **Question 3.10.** Is every (preserved) extreme point of $B_{X\widehat{\otimes}_\pi Y}$ a basic tensor?



isomorphic (it follows for instance from [19, Proposition 2.3]) and therefore $((-e_1 + e_n) \otimes (-e_1 + e_{k_n}))_n$ is not weakly convergent to $e_1 \otimes e_1$.

We end the paper with some open questions.

Question 3.9. *Is $x \otimes y$ a preserved extreme point of $B_{X\widehat{\otimes}_\pi Y}$ whenever x and y are preserved extreme points of B_X and B_Y?*

Question 3.10. *Is every (preserved) extreme point of $B_{X\widehat{\otimes}_\pi Y}$ a basic tensor?*

References

[1] R. J. Aliaga, *Extreme points in Lipschitz-free spaces over compact metric spaces*, Mediterr. J. Math. **19** (2022), article 32.
[2] R. J. Aliaga and A. J. Guirao, *On the preserved extremal structure of Lipschitz-*

Figure 1: The two open questions in arXiv:2211.13559, PDF page 13.

Throughout, all spaces are real. A point $u \in B_E$ is *preserved extreme* when its canonical image is an extreme point of $B_{E^{**}}$. A point $x \in S_X$ is *strongly exposed* by $x^* \in S_{X^*}$ when $x^*(x) = 1$ and $x^*(x_\alpha) \to 1$ for a net in B_X forces $\|x_\alpha - x\| \to 0$. The sequential and net formulations agree here because failure of the uniform modulus below produces a counterexample sequence.

2 Proof intuition

Suppose x^* strongly exposes x and y^* norms y. If an elementary tensor $a \otimes b$ nearly attains the tensor functional $x^* \otimes y^*$, then $x^*(a)$ and $y^*(b)$ have the same sign and both have modulus close to one. Multiplying both a and b by that sign does not change the tensor, and strong exposure forces the adjusted first factor close to x. Thus every near-maximizer lies close to the tensor fiber $x \otimes B_Y$. Convexity gives the same conclusion for every near-maximizer in the whole projective tensor ball.

Goldstine's theorem then shows that the norm-one face of $B_{(X\widehat{\otimes}_\pi Y)^{**}}$ exposed by $x^* \otimes y^*$ is contained in $x \otimes B_{Y^{**}}$. Any bidual midpoint decomposition of $x \otimes y$ therefore becomes a midpoint decomposition of y in $B_{Y^{**}}$, which is trivial when y is preserved extreme.

For Question 3.10, a norm-attaining nuclear representation is already a convex-series representation by normalized basic tensors. Extremality forces every term with positive weight to equal the tensor itself.

3 A tensor-face concentration lemma

For $x \in S_X$, write

$$M_x : Y \longrightarrow X\widehat{\otimes}_\pi Y, \qquad M_x(v) = x \otimes v.$$

This is a linear isometry.

Lemma 1 (Concentration of a strongly exposed tensor face)**.** *Let $x \in S_X$ be strongly exposed by $x^* \in S_{X^*}$, let $y^* \in S_{Y^*}$, and put $\Phi = x^* \otimes y^* \in (X\widehat{\otimes}_\pi Y)^*$. Then*

$$\{z^{**} \in B_{(X\widehat{\otimes}_\pi Y)^{**}} : \Phi(z^{**}) = 1\} \subseteq M_x^{**}(B_{Y^{**}}).$$

Moreover, if $z^{**} = M_x^{**}(v^{**})$ belongs to this face, then $y^*(v^{**}) = 1$.

Proof. For $0 < \delta < 1$, set

$$\omega(\delta) = \sup\{\|a - x\| : a \in B_X,\ x^*(a) > 1 - \delta\}.$$

Strong exposure gives $\omega(\delta) \to 0$ as $\delta \downarrow 0$.

Consider $a \in B_X$, $b \in B_Y$ such that

$$x^*(a)y^*(b) > 1 - \delta.$$

The two scalar factors have the same sign and both have modulus greater than $1-\delta$. If $\sigma \in \{-1, 1\}$ is their common sign, then $x^*(\sigma a) > 1 - \delta$ and

$$\|a \otimes b - x \otimes (\sigma b)\|_\pi = \|(\sigma a) \otimes (\sigma b) - x \otimes (\sigma b)\|_\pi \le \omega(\delta). \tag{1}$$

We next record the convex consequence of (1). If (w_α) is any net in $B_{X\widehat{\otimes}_\pi Y}$ with $\Phi(w_\alpha) \to 1$, then

$$\operatorname{dist}(w_\alpha, M_x(B_Y)) \longrightarrow 0. \tag{2}$$

Indeed, approximate each w_α in norm by a finite convex combination of elements of $B_X \otimes B_Y$, with approximation error tending to zero and with its Φ-value still tending to one. Fix $\delta > 0$ and call a term good when its Φ-value is greater than $1 - \delta$. Since every term has Φ-value at most one, the total weight of the bad terms tends to zero. Replace every good term, using (1), by a point of $M_x(B_Y)$, and replace the bad part by zero. The replacement of the good part is within $\omega(\delta)$, while the bad part has norm tending to zero. Hence

$$\limsup_\alpha \operatorname{dist}(w_\alpha, M_x(B_Y)) \le \omega(\delta).$$

Letting $\delta \downarrow 0$ proves (2).

Now take z^{**} in the face in the statement. By Goldstine's theorem, there is a net $(w_\alpha) \subset B_{X\widehat{\otimes}_\pi Y}$ converging weak-star to z^{**}. Then $\Phi(w_\alpha) \to 1$, so by (2) choose $v_\alpha \in B_Y$ with

$$\|w_\alpha - M_x(v_\alpha)\| \longrightarrow 0.$$

After passing to a subnet, v_α converges weak-star in $B_{Y^{**}}$ to some v^{**}. It follows that $z^{**} = M_x^{**}(v^{**})$. Finally,

$$1 = \Phi(z^{**}) = y^*(v^{**}),$$

which proves the last assertion. □

4 A positive subcase of Question 3.9

Theorem 2. *Let X, Y be Banach spaces. If $x \in S_X$ is strongly exposed in B_X and $y \in S_Y$ is a preserved extreme point of B_Y, then $x \otimes y$ is a preserved extreme point of $B_{X\widehat{\otimes}_\pi Y}$. The same conclusion holds with the roles of X and Y reversed.*

Proof. Let $x^* \in S_{X^*}$ strongly expose x, and choose $y^* \in S_{Y^*}$ with $y^*(y) = 1$. Suppose

$$x \otimes y = \frac{z_1^{**} + z_2^{**}}{2}, \qquad z_1^{**}, z_2^{**} \in B_{(X\widehat{\otimes}_\pi Y)^{**}}. \tag{3}$$

Put $\Phi = x^* \otimes y^*$. Since $\Phi(x \otimes y) = 1$ and $\Phi(z_j^{**}) \le 1$, equality in (3) forces $\Phi(z_1^{**}) = \Phi(z_2^{**}) = 1$. Lemma 1 gives

$$z_j^{**} = M_x^{**}(v_j^{**}) \quad \text{for some } v_j^{**} \in B_{Y^{**}}.$$

Because M_x^{**} is an isometry and hence injective, (3) yields

$$y = \frac{v_1^{**} + v_2^{**}}{2}.$$

The point y is extreme in $B_{Y^{**}}$, so $v_1^{**} = v_2^{**} = y$. Therefore $z_1^{**} = z_2^{**} = x \otimes y$, proving that $x \otimes y$ is preserved extreme. Symmetry of the projective tensor product gives the reversed version. □

No approximation property and no assumption on compactness of operators $X \to Y^*$ is used. This is complementary to the later compact-operator subcase in [2].

5 Norm-attaining extreme tensors are basic

Call $z \in X\widehat{\otimes}_\pi Y$ *projective-norm-attaining* when it admits a norm-convergent representation

$$z = \sum_{n=1}^{\infty} x_n \otimes y_n, \qquad \sum_{n=1}^{\infty} \|x_n\|\,\|y_n\| = \|z\|_\pi. \tag{4}$$

Theorem 3. *Every projective-norm-attaining extreme point of $B_{X\widehat{\otimes}_\pi Y}$ is a basic tensor. In fact, if $z \in \operatorname{ext} B_{X\widehat{\otimes}_\pi Y}$ and (4) holds, then $z = u \otimes v$ for some $u \in S_X$ and $v \in S_Y$.*

Proof. An extreme point of a nontrivial closed unit ball has norm one. Remove zero terms from (4) and put

$$\lambda_n = \|x_n\|\,\|y_n\|, \quad u_n = \frac{x_n}{\|x_n\|}, \quad v_n = \frac{y_n}{\|y_n\|}.$$

Then $\lambda_n > 0$, $\sum_n \lambda_n = 1$, and

$$z = \sum_n \lambda_n (u_n \otimes v_n), \qquad u_n \otimes v_n \in B_{X \widehat{\otimes}_\pi Y}. \tag{5}$$

Fix n. If $\lambda_n = 1$, (5) already makes z basic. Otherwise write

$$z = \lambda_n (u_n \otimes v_n) + (1 - \lambda_n) w_n, \qquad w_n = \frac{1}{1 - \lambda_n} \sum_{k \neq n} \lambda_k (u_k \otimes v_k) \in B_{X \widehat{\otimes}_\pi Y}.$$

Extremality of z forces $u_n \otimes v_n = z$. Choosing any n proves the claim. □

Corollary 4. *Any non-basic extreme point, and hence any non-basic preserved extreme point, whose existence is contemplated in Question 3.10 must fail to attain its projective norm in the sense of (4).*

6 Limitations and novelty check

Theorem 2 does not cover a pair of preserved extreme points when neither point is strongly exposed. Theorem 3 does not exclude non-basic extreme points arising only in the completion through non-norm-attaining representations. Thus neither source question is claimed solved in full.

A bounded arXiv search used the exact question phrases and the terms "preserved extreme", "strongly exposed", "denting", "basic tensor", and "projective tensor". The relevant later paper [2] proves an affirmative version of Question 3.9 under $L(X, Y^*) = K(X, Y^*)$ and an approximation-property hypothesis, and explicitly describes the general question as open. No exact source for the two results in this packet was located. This supports, but does not certify, novelty.

7 Verification and human-review focus

The source question was checked against arXiv PDF page 13. The proof was checked at the three sensitive points: simultaneous sign correction in (1), vanishing bad mass in (2), and identification of the bidual face through Goldstine's theorem. No numerical computation is involved.

Human review should focus first on Lemma 1, especially the passage from finite convex combinations to the weak-star bidual face. If that lemma is accepted, Theorem 2 is immediate.

References

[1] L. C. García-Lirola, G. Grelier, G. Martínez-Cervantes, and A. Rueda Zoca, *Extremal structure of projective tensor products*, Results Math. 78 (2023), article 196; arXiv:2211.13559.

[2] R. J. Aliaga, L. C. García-Lirola, J. Guerrero-Viu, M. Raja, and A. Rueda Zoca, *Almost preserved extreme points*, arXiv:2510.21234, 2025.

[3] R. A. Ryan, *Introduction to Tensor Products of Banach Spaces*, Springer Monographs in Mathematics, Springer, 2002.